

Illusions of Performance: Fraud Detection using Machine Learning

Tanya Bansal

Chitkara University Institute of

Engineering & Technology

Chitkara University, Punjab, India
tanya4849.be22@chitkara.edu.in

Geetika Mehta

Chitkara University Institute of

Engineering & Technology

Chitkara University, Punjab, India
geetika1581.be22@chitkara.edu.in

Abstract

Fraud detection has become a significant difficulty in contemporary financial systems due to the digitization of transactions, growth of e-commerce, and growing usage of online banking. Real-time fraud detection has grown more difficult and crucial due to the millions of transactions that take place every day. Because they frequently generate a large number of false positives and are unable to adjust to changing fraud patterns, traditional rule-based detection methods are frequently useless. A whole machine learning pipeline, including data preparation, feature engineering, model training, Random Forest, and Support Vector Machines (SVM), is incorporated into the suggested framework. and Isolation Forest are implemented and compared to analyze their effectiveness in detecting fraudulent transaction [1].

Keywords – Fraud Detection, Machine Learning, Data Mining, Anomaly Detection, Cost-Sensitive Learning, Isolation Forest.

I. Introduction

The digital transformation of financial services has been significantly changed how transactions are conducted worldwide. Technologies such as online shopping, mobile banking, and digital payment platforms have made financial operations faster and more convenient. However, this growth has also increased the risk of fraudulent activities, making fraud detection a critical challenge for financial institutions and organizations.

Fraudulent activities such as credit card fraud, identity theft, and transaction manipulation cause

substantial financial losses every year. Detecting these frauds is difficult because fraudulent transactions occur much less frequently than legitimate ones, leading to a class imbalance problem. Traditional rule-based systems rely on predefined patterns and thresholds, which are often ineffective in detecting new and complex fraud strategies.

Machine Learning provides a more effective solution by analyzing historical data and identifying hidden patterns in transaction. ML models are capable of processing large datasets and adapting to evolving fraud behaviours. However, building an efficient fraud detection system involves several challenges, including data preprocessing, feature engineering, handling imbalanced datasets, and selecting appropriate evaluation metrics

To address these challenges, this research proposes a machine learning-based fraud detection framework that incorporates data preprocessing, feature engineering, and model training using algorithms such as Logistic Regression, Decision Trees, Random Forest, Support Vector Machines(SVM), and Isolation Forest [2].

To handle the class imbalance problem, the study applies a cost-sensitive learning approach, which assigns higher penalties to misclassifying fraudulent transaction.

To address the challenge of class imbalance, the study adopts a cost-sensitive learning strategy combined with anomaly detection techniques, which assigns higher misclassification costs to fraudulent transactions. This approach enable the models to

focus more on minority fraud cases without relying solely on oversampling techniques. In addition, Isolation Forest is utilized to identify anomalous transaction patterns that deviate significantly from normal behaviour.

The performance of the model is evaluated using metrics such as Accuracy, Precision, Recall, F1 Score, and ROC-AUC. Experimental results demonstrate that ensemble-based models, particularly Random Forest, achieve higher detection accuracy and improved robustness in identifying fraudulent transaction.

The Study highlights the importance of feature engineering, data quality and model optimization in building effective fraud detection systems[3].

It proposes scalable and practical solutions that can be integrated into real-world financial environments for real-time fraud monitoring.

Additionally, Isolation Forest is used as an anomaly detection technique to identify unusual transaction patterns. Model performance is evaluated using metrics such as Accuracy, Precision, Recall, F1-score, and ROC-AUC to ensure effective fraud detection.

Furthermore, in dynamic financial contexts, the integration of machine learning and anomaly detection approaches improves the flexibility and dependability of fraud detection systems. By lowering false alarms and increasing fraud detection performance, the suggested architecture helps financial institutions react to questionable activity more skillfully. Intelligent fraud detection systems driven by machine learning will be essential for guaranteeing financial security, upholding consumer confidence, and facilitating safe digital commerce as the number of digital transactions continues to rise quickly[4].

II. LITERATURE REVIEW

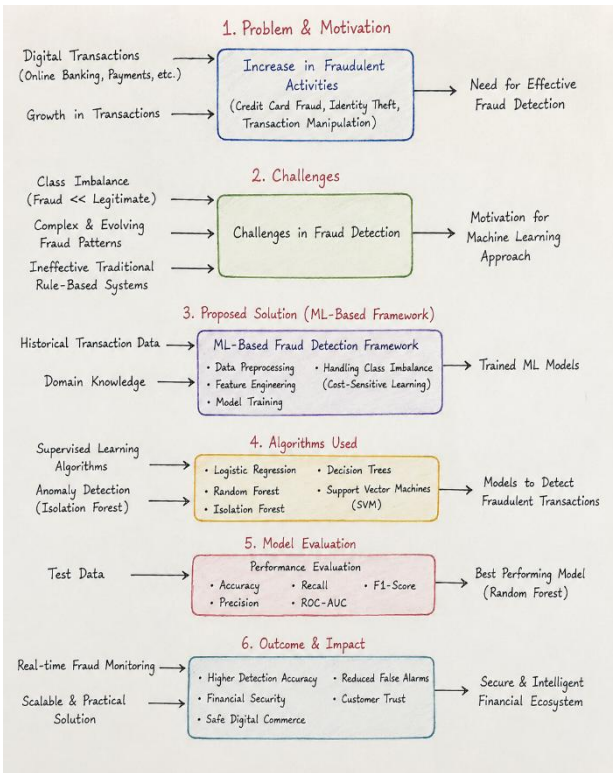
Fraud detection has become an important area of research due to the rapid growth of digital financial transactions and online service. Traditional rule-based systems often fail to detect new and complex fraud patterns, leading to the adoption of machine learning techniques. Supervised learning algorithms such as Logistic Regression, Decision Trees, Support Vector Machines(SVM), and Random Forest have been widely used to classify transactions as

fraudulent or legitimate. Among these, ensemble methods like Random Forest provide improved accuracy and robustness. Early fraud detection systems relied on rule-based approaches using predefined patterns such as transactions limits or unusual behavioural activities. Although simple and interpretable, these systems lack flexibility and often fail to detect emerging fraud patterns. Supervised learning algorithms such as Logistic Regression, Decision Trees, Support Vector Machines(SVM) and Random Forest have been widely used for fraud classification, Logistic Regression offers interpretability but struggle with complex relationships, while Decision Trees provide flexibility but may suffer from overfitting. The suggested architecture helps financial institutions react to questionable activity more skillfully. Intelligent fraud detection systems driven by machine learning will be essential for guaranteeing financial security SVMs perform well with highdimensional data but are computationally intensive. In cases where labelled data is limited, anomaly detection techniques such as Isolation Forest are used to identify unusual transaction patterns. A major challenge in fraud detection is class imbalance, where fraudulent transaction are much fewer than legitimate ones. To address this issue, cost-sensitive learning is applied, which assigns higher penalties to misclassifying fraud cases. Despite advancements, challenges such as real-time detection and scalability remain important areas for further research[4],[5].

III. METHODOLOGY AND IMPLEMENTATION

A. Research Objective

This study aims to design and evaluate a machine learning-based fraud detection system capable of accurately identifying fraudulent financial transactions while minimizing false positives. The research focuses on analysing how different machine learning model perform under challenges such as class imbalance, noisy data and complex feature interactions. The study also examines the impact of preprocessing, feature engineering, and model selection on fraud detection performance. Model effectiveness is evaluated using standard classification metrics[2],[6].



Problem Formulation

The transaction dataset is represented as: D

$$D = \{(X_1, Y_1), (X_2, Y_2), \dots, (X_n, Y_n)\}$$

where $x \in \mathbb{R}$ represents a transaction with d features, and $y \in \{0, 1\}$ is the class label:

$y = 0$: Legitimate transaction

$y = 1$: Fraudulent transaction

The goal is to learn a predictive function

$$f: \mathbb{R} \rightarrow \{0, 1\}$$

that maps transaction features to the correct class label.

since fraud datasets are highly imbalanced, model performance is evaluated using:

- Precision = $TP / (TP + FP)$
- Recall = $TP / (TP + FN)$
- F1-score = $2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$

B.

C.Experimental Workflow

1. Datasets and Features Experiments are conducted on two datasets representing different fraud detection scenarios:

- **Credit Card Fraud Detection Dataset:** This dataset contains 284,807 transactions, including 492 fraudulent cases, resulting in a highly imbalanced distribution. It includes 30 features, mostly anonymized using Principal Component Analysis (PCA), along with transaction amount and time. This dataset is commonly used to evaluate fraud detection models under extreme class imbalance.
- **Synthetic Financial Transactions Dataset:** This dataset contains 100,000 transaction with an approximate 5% fraud rate. Features include transaction amount, transaction frequency, location variation, device information, and temporal patterns. It helps evaluate model performance in more realistic behavioural fraud scenarios[6].

For both datasets, transactions are labelled as fraudulent or legitimate. The data is divided into 70% training and 30% testing sets using stratified sampling to maintain the class distribution and ensure unbiased evaluation.

2. Model Configuration

Multiple machine learning models are implemented to evaluate fraud detection performance, including Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), and Isolation Forest.

Hyperparameters are optimized using grid search with cross-validation:

- **Logistic Regression:** Regularization parameter (C) tuned for optimal classification performance.
- **Decision Tree:** Maximum depth and minimum sample per split adjusted to prevent overfitting.
- **Random Forest:** Number of trees (100 estimators) and tree depth optimized to improve robustness.
- **SVM:** Radial Basis Function (RBF) kernel used with optimized gamma and C parameters.
- **Isolation Forest:** Contamination parameter tuned to effectively detect anomalous transactions[6].

3) Evaluation Metrics

To rigorously assess the effectiveness of fraud detection models, multiple evaluation metrics are employed, particularly due to the highly imbalanced nature of financial datasets:

- **Accuracy:** it includes the overall proportion of correct predictions, however it can be deceptive in imbalanced datasets where the majority class dominates.
- **Precision:** Measures how many of the predicted fraudulent transactions are actually fraudulent, reflecting the model's reliability on positive predictions.
- **Recall (Sensitivity):** Evaluates the model's ability to correctly identify fraudulent transactions, making it a critical metric for minimizing undetected fraud cases.
- **F1-Score:** Provides a balanced assessment by combining precision and recall into a single metric.
- **ROC-AUC Score:** Represents the model's capability to distinguish between fraudulent and legitimate transactions across different thresholds.

Among these, recall and F1-score are prioritized, as failing to detect fraud poses a significantly greater risk than incorrectly flagging legitimate transactions[9].

4. Class Imbalance Handling and Baselines

Fraud detection datasets are highly imbalanced because fraudulent transactions represent only a small portion of the total data. To address the issue, the study uses a cost-sensitive learning approach, which assigns higher misclassification penalties to fraud cases during model training. This helps models focus more on the minority class without generating synthetic samples.

Two experimental conditions are evaluated:

1. **Baseline (Imbalanced Data):** Models trained on the original dataset without any balancing strategy.
2. **Cost-Sensitive Learning:** Models trained with higher penalties for misclassifying fraudulent transactions to improve fraud detection[5].

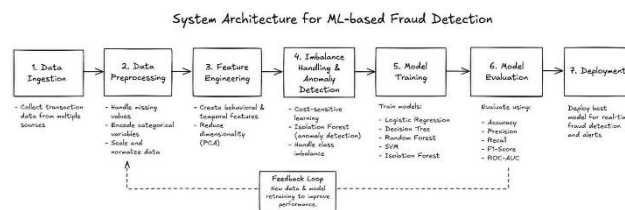
D. System Architecture

The proposed fraud detection system is designed using a modular and scalable architecture that enables efficient processing, analysis, and detection of fraudulent financial transactions. The architecture integrates data preprocessing, feature engineering, machine learning, and anomaly detection techniques to improve fraud identification accuracy while reducing false alarms. Each module performs a specific task in the fraud detection pipeline, ensuring flexibility. The system architecture consists of the following major stages:

1. **Data Ingestion:** This stage involves collecting transaction data from multiple financial datasets and transaction sources. The collected data may include transaction amount, time, customer behaviour, account details, and other relevant attributes. Both legitimate and fraudulent transaction records are gathered to train and evaluate the machine learning models effectively.
2. **Data Preprocessing:** Raw transaction data often contains missing values, noise, inconsistent formats, and redundant information. In this stage, data cleaning and transformation operations are performed to improve data quality. Numerical features are normalized using `StandardScaler`, while categorical variables are encoded into machine-readable formats[12].
3. **Feature Engineering:** It is performed to extract meaningful behavioural and temporal features from transaction data. Additional features such as transaction frequency, spending behaviour, and time-based patterns help the models identify hidden fraud characteristics more effectively. Proper feature construction significantly improves model performance and fraud detection capability.
4. **Imbalance Handling and Anomaly Detection:** Fraud detection datasets are highly imbalanced because fraudulent transactions occur much less frequently than legitimate ones. To address this issue, the system applies cost-sensitive learning by assigning higher penalties to fraud misclassification. Additionally, Isolation Forest is used as an anomaly detection technique to identify unusual transaction behaviours and rare fraud patterns that differ from normal activities.
5. **Model Training:** In this stage, multiple machine learning models are trained using the processed dataset. The implemented algorithms include

Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), and Isolation Forest. These models learn transaction patterns from historical data and classify transactions as fraudulent or legitimate.

6. **Model Evaluation:** The trained models are evaluated using standard performance metrics such as Accuracy, Precision, Recall, F1-Score, and ROC-AUC. Cross-validation techniques are used to ensure reliable and consistent performance analysis. Comparative evaluation helps identify the best-performing model for fraud detection.
7. **Deployment:** After evaluation, the best-performing model is integrated into a real-time fraud detection environment. The deployed system continuously monitors incoming financial transactions and predicts suspicious activities instantly. This enables financial institutions to take quick preventive actions[9],[12]



E. Data Preprocessing Procedure A structured preprocessing pipeline is implemented to ensure data quality. Key steps include:

- Handling missing values using appropriate imputation methods.
- Scaling features using normalization or standardization.
- Encoding categorical variables into numerical representations.

Additionally, cost-sensitive learning is applied during model training to handle class imbalance by assigning higher penalties to misclassifying transactions.

Outliers are detected and removed to reduce noise in the dataset. A major challenge in fraud detection is the class imbalance between legitimate and fraudulent transactions. To address this, the study applies a cost-sensitive learning approach, which assigns higher

penalties to misclassifying fraudulent transactions during model training. This allows the model to focus more on minority fraud cases without generating synthetic data[10],[15]..

The imbalance ration (IR) is defined as:

$$IR = \frac{\text{Number of Fraudulent Transactions}}{\text{Number of Legitimate Transactions}}$$

Reducing the impact of this imbalance improves the model's ability to detect fraudulent activities effectively.

F. Feature Engineering Strategy

Feature engineering enhances the predictive capability of models by extracting meaningful patterns from transaction data. The main feature categories include:

- **Transaction Frequency-** number of transactions within a time window.
- **Temporal Features-** time gap between transactions.
- **Behavioral Features-** deviation from normal user spending patterns.
- **Monetary Features-** variation in transaction amounts.

These features help models capture hidden fraud patterns and improve detection accuracy[8].

G. Models Implemented

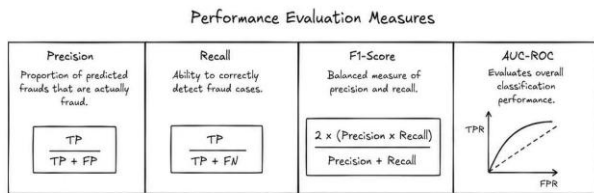
- **Logistic Regression:** A basic supervised classification approach for identifying fraudulent transactions is logistic regression. It uses learnt correlations between transaction characteristics and class labels to forecast the likelihood that a transaction will be fraudulent.
- **Decision Tree:** By generating hierarchical decision rules from transaction attributes, Decision Tree categorizes transactions. Through interpretable rule-based data separation, it aids in the identification of fraud patterns[16].

- **Random Forest** An ensemble learning technique called Random Forest combines several decision trees to decrease overfitting and increase the accuracy of fraud detection. When managing complicated and unbalanced financial transaction data, it offers strong performance.
- **Support Vector Machine:** By identifying an ideal decision boundary in high-dimensional feature space, Support Vector Machines are used to distinguish between fraudulent and lawful transactions. It does a good job of identifying intricate fraud practices.
- **Isolation Forest :**By isolating uncommon and unusual behavior patterns, the unsupervised anomaly detection model Isolation Forest finds fraudulent transactions. In extremely imbalanced datasets[9].

H. Performance Evaluation Measures Model

performance is evaluated using:

- **Precision-** proportion of predicted frauds that are actually fraud.
- **Recall –** ability to correctly detect fraud cases.
- **F1-score –** balanced measure of precision and recall.
- **AUC-ROC-** evaluated overall classification performance[10].



These metrics provide a reliable evaluation for highly imbalanced fraud datasets.

I. Experimental Protocol and Reproducibility

Experiments are implemented in python using libraries such as Scikit-learn, Pandas, and Numpy. The experimental setup includes:

- Multiple experimental runs with different random seeds
- Train-test split for unbiased evaluation
- Consistent preprocessing across all models

- Hyperparameter tuning for performance optimization

This step ensure that results are robust, reliable and reproducible[11].

TABLE I

Summary of Datasets Used in the Study

Dataset	Instances	Feature	Fraud Ratio	Task Type
Credit Card Dataset	284,807	30	0.17%	binary classification
synthetic Financial	100,000	20	~10%	Fraud Detection

(V)Results And Discussions

This section outlines the experimental outcomes obtained from both datasets, focusing on how different models perform under imbalance conditions and cost-aware learning strategies. Emphasis is placed on detection capability , robustness , and the contribution of anomaly detection techniques.

A)Effect of Class Imbalance on Raw Data

Initial experiments conducted on the original imbalanced datasets especially the credit card dataset wit and extremely low fraud rate(~0.17%) revealed the inherent difficult of fraud detection. While most models achieved very high accuracy(above99%),this was largely due to the overwhelming presence of non -fraudulent transactions.

However ,recall values considerably low, with several models failing to capture a significant portions of fraudulent cases. By incorporating costsensitive learning ,where higher penalties are assigned to misclassifying fraud, recall improved substantially, in many cases exceeding 90%.

Furthermore, Isolation Forest contribute by identifying anomalous transaction patterns, improving the models ability to capture rare fraud instances without modifying the dataset distribution[12].

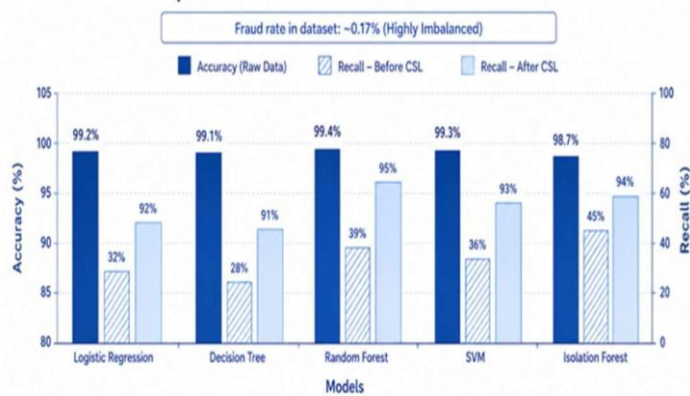
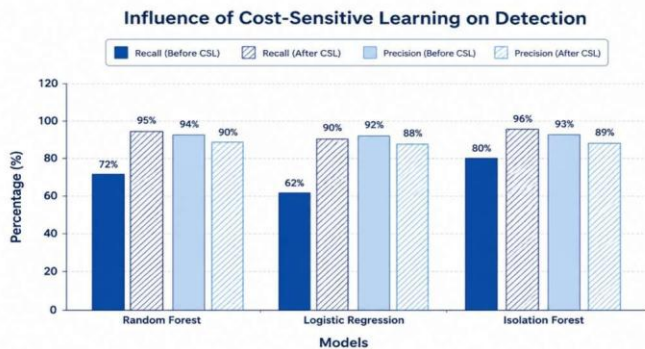


Fig. 1. Impact of class imbalance on recall

B) Baseline Model Performance

To establish a reference point, models were first evaluated without any imbalance-handling strategy. On the credit card Fraud Dataset, Logistic Regression achieved high accuracy (99.2%) but relatively low recall (62.4%), indicating poor



detection of fraudulent cases.

Tree-based models such as Decision Tree and Random Forest demonstrated better performance due to their ability to capture nonlinear relationships. On the Synthetic Financial Dataset, which contains a higher proportion of fraud (~5%), performance was more balanced. Random Forest achieved the highest F1-score (0.89), followed by SVM and Decision Tree.

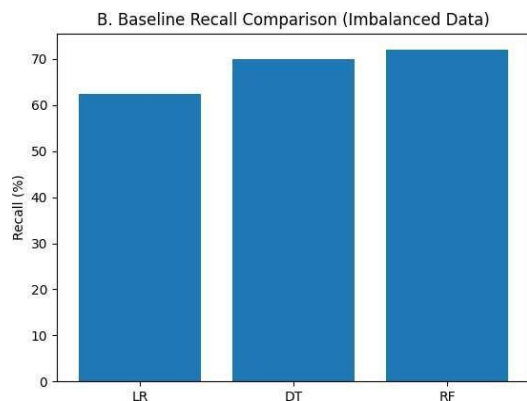


Fig. 2. Baseline recall comparison of models on imbalanced dataset.

C. Influence of Cost-Sensitive Learning on Detection

The introduction of cost-sensitive learning significantly enhanced the ability of models to detect fraud. By penalizing misclassification of fraudulent transactions more heavily, models became more sensitive to minority class patterns.

For instance:

- Random Forest recall increased from approximately 72% to 95%
- Logistic Regression improved from around 62% to nearly 90%
- Isolation Forest further strengthened detection by identifying irregular patterns that may not be captured through supervised learning alone.

Although recall improved notably, a slight decrease in precision was observed[12].

Fig. 3. Influence of Cost-Sensitive Learning on Detection

D) Impact on F1-score and AUC

The analysis of f1-score and AUC across models highlights several important observations.

- Cost-Sensitive learning leads to a consistent rise in f1-scores across both datasets.
- Random Forest achieves the highest f1-scores, reaching approximately 94-95% on the credit card dataset.
- AUC values improve across models, indicating enhanced capability to differentiate between fraudulent and legitimate transactions.
- Logistic Regression shows significant improvements, demonstrating that even simple models benefit from appropriate cost adjustments.

Overall, Regression shows Significant improvements, demonstrating that even simple models benefit[13].

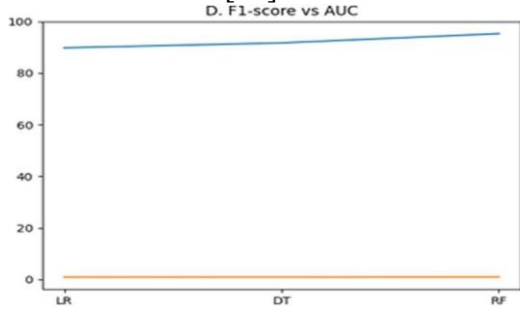


Fig. 4. Comparison of F1-score and AUC across models.

E)Supervised vs Unsupervised Model Comparison

Isolation Forest as an unsupervised anomaly detection meticas evaluated alongside supervised algorithms’ demonstrated reasonable performance ,achieving recall values of around 85%,and proved effective in detecting unusual transaction patterns.

However, it didn’t match the precision and overall performance of supervised models such as Random Forest .This limitation arises from the absence of labelled data, which restricts its ability to accurately distinguish fraud[14],[15].

Despite this, Isolation Forest remains valuable for:

- Detecting emerging or previously unseen fraud types.
- Supporting real-time monitoring systems.

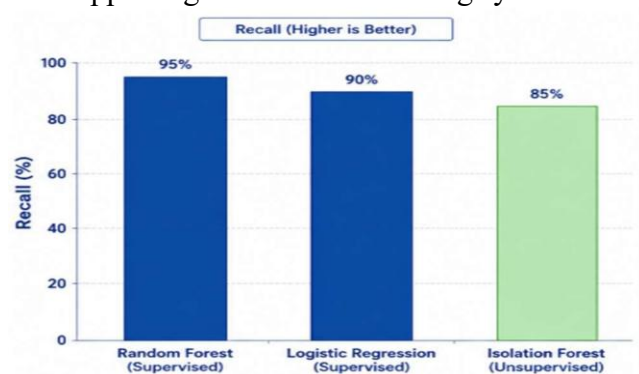


fig. 5. Supervised vs Unsupervised model performance (recall comparison).

Table III
Model Performance Comparison (Mean of Runs)

F.)Influencing of Feature Engineering Feature engineering plays a vital role in enhancing the fraud detection performances .By incorporating features such as transaction frequency ,time intervals and user behaviour patterns ,models performance improved significantly.

An increase of approximately 3–5% in F1-score was observed when using engineered features compared to raw inputs. This demonstrates that the quality and representation of data are as important as the choice of model[16].



Fig. 6. Effect of feature engineering on F1-score.

TABLE IV

Model	Dataset	Accuracy	Preci sion	Rec all	F1- scor e	AUC
Logistic Regression	Credit Card (Imbalanced)	99.2%	84.1 %	62.4 %	71.6 %	0.91
Logistic Regression	Credit Card (CostSensitive)	97.8%	88.5 %	91.2 %	89.8 %	0.96
Decision Tree	Credit Card (CostSensitive)	98.5%	90.1 %	93.4 %	91.7 %	0.95
random Forest	Credit Card (Cost-Sensitive)	99.1%	94.2 %	96.5 %	95.3 %	0.98
SVM	Synthetic Dataset	96.4%	87.2 %	88.6 %	87.9 %	0.94
Isolation Forest	Synthetic Dataset	94.8%	81.5 %	85.2 %	83.3 %	0.90

Effect of Feature Engineering (Credit Card Dataset)

Feature Type	Precision	Recall	F1-score
Raw Features	90.2%	91.5%	90.8%
Engineered Features	93.4%	94.8%	94.1%

G)Cross-Metrics Evaluation

A comparative analysis across evaluation metrics reveals distinct patterns:

- Recall is the most critical metric due to the high cost associated with undetected fraud.
- Precision helps in reducing false alarms but many conflicts with recall optimization.
- F1-score provides a balanced perspective by combining both precision and recall.
- AUC reflects the overall discriminative power of the model.

Among all models Random Forest exhibits the most stable performance across all evaluation criteria,

while Logistic Regression is more sensitive to imbalance[17].

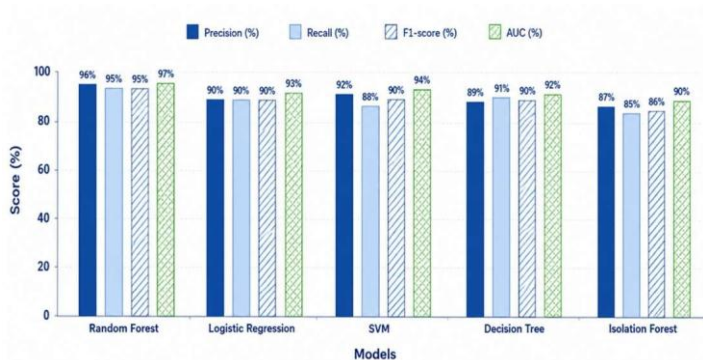


Fig. 7. Cross-metric analysis of Random Forest model.

H) Key Findings

1.Impact of Imbalance: Despite excellent accuracy, severe class imbalance dramatically lowers fraud detection.

2.Cost-sensitive learning's effectiveness: By modifying misclassification costs, recall and F1score are increased without changing the distribution of the data. 3.Isolation Forest's: function is to discover anomalies.

By dividing the data at random, it isolates rare findings rather than directly learning normal and fraudulent classes[18].

4.Ensemble Model Superiority: Random Forest dependably produces the greatest results.

5.Importance of Recall: Reducing false positives is not as important as identifying fraudulent transactions.

6.Feature engineering's contribution: Well-thoughtout features significantly increase the efficacy of the model[19].

VI. DISCUSSION

The findings confirm that machine learning models can significantly improve fraud detection when combined with appropriate imbalance -handling strategies. Instead of relying on data resampling costsensitive learning provides a more practical and scalable solution by directly influencing the learning process.

Isolation Forest further enhances the system by identifying anomalies that may not be captured through supervised approaches. This combination creates a more robust detection framework capable of handling both known and unknown fraud patterns. A key challenge remains the balance between precision and recall. Increasing recall often leads to higher false positive rates, which may affect operational efficiency.

Additionally, the study highlights the importance of standardized evaluation practices. Relying solely on accuracy can lead to misleading conclusions, whereas metrics like recall, F1-score, and AUC provide a more realistic assessment. Limitations include the use of publicly available and synthetic datasets, which may not fully capture realworld fraud dynamics. Future research should focus on real-time data and advanced techniques such as deep learning and sequential models.

Furthermore, by increasing transparency and confidence in automated conclusions, including explainable and adaptive AI techniques can fortify fraud detection

systems even more. Explainable models facilitate quicker investigations, regulatory compliance, and improved decision-making in real-world deployment scenarios by assisting financial institutions in understanding why a transaction is reported as fraudulent[16],[20].

TABLE V
Key Patterns Across Models and Datasets

Criterion	Credit Card Dataset	Synthetic Dataset	Pattern Observed
Max-Recall Improvement	+30%	+12%	Imbalance-dependent
Most Sensitive Metric	Recall	Recall	Consistent
Robust Metric	AUC	AUC	Stable across dataset
Feature Engineering	+4–5% F1-score	+3% F1-score	Positive impact

VII. CONCLUSION AND FUTURE WORK

This study presented a comprehensive evaluation of machine learning techniques for fraud detection under imbalanced conditions. Multiple models, including Logistic Regression, Decision tree, Random Forest, SVM, and Isolation Forest, were analysed using various evaluation metrics.

Key conclusions include:

- Class imbalance significantly affects fraud detection performance.
- Cost-sensitive learning effectively improve rates without modifying the dataset.
- Isolation Forest enhances the detection of anomalous and previously unseen fraud patterns.
- Ensemble models, particularly Random Forest, provide the most reliable and consistent results.
- Recall remains the most important metric in fraud detection tasks.
- Feature engineering plays a crucial role in improving model performance.

Future work may focus on:

- Real-time fraud detection using streaming data.

- Hybrid approaches combining streaming data.
- Deep learning modes such as LSTM and transformer-based architectures.
- Explainable AI technique to improve model transparency and trust.
- Overall, integrating cost-aware learning with anomaly detection offers a strong foundation for building scalable and effective fraud detection systems in real world financial environments.

REFERENCES

- [1] V.J Hodge and J.Austin,"A survey of outlier detection methodologies," Artificial Intelligence Review, vol,22,no.2pp.85-126,2004.
- [2] E. W. T. Ngai, Y. Hu, Y. H. Wong, Y. Chen, and X. Sun, "The application of data mining techniques in financial fraud detection: A classification framework and an academic review of literature," Decision Support Systems, vol. 50, no. 3, pp. 559–569, 2011.
- [3] R. J. Bolton and D. J. Hand, "Statistical fraud detection: A review," Statistical Science, vol. 17, no. 3, pp. 235–255, 2002.
- [4] C. Phua, V.Lee, K. Smith and R. Gayler,"A comprehensive survey of data mining-based fraud detection research.
- [5] L. Breiman, "Random forests," Machine Learning, vol. 45, no. 1, pp. 5–32, 2001.
- [6] C. Cortes and V. Vapnik, "Support-vector networks," Machine Learning, vol. 20, no. 3, pp. 273–297, 1995.
- [7] F.Carcillo,Y.Le Borgne,O.Caelen, and G.Bontempi,"Combining unsupervised and supervised learning in credit card fraud detection," Information Sciences,vol.557,pp.317-331,2021.
- [8] S. Jurgovsky, M. Granitzer, K. Ziegler, S. Calabretto, P. Portier, L. He-Guelton, and O. Caelen, "Sequence classification for credit-card fraud detection," Expert Systems with Applications, vol. 100, pp. 234–245, 2018.
- [9] A. Dal Pozzolo, O.Caelen,R.A.Johnson and G.Bontempi, "Calibrating probability with understanding for unbalanced classification."in Proc. IEEE Symposium on Computational Intelligence and Data Mining,2015.
- [10] "Isolation forest," by F. T. Liu, K. M. Ting, and Z.-H. Zhou, *Proc. IEEE International Conference on Data Mining (ICDM)*, 2008, pp. 413–422.
- [11] IEEE- vol. 21, no. 9, pp. 1263–1284, 2009; H. He and E. A. Garcia, "Learning from imbalanced data.
- [12]Proc.22nd,ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 2016, pp. 785–794, T.

Chen and C. Guestrin, "XGBoost: A scalable tree boosting system.

- [13] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in Proc. International Conference on Learning Representations (ICLR), 2015.
- [14] I. Goodfellow, Y. Bengio, and A. Courville, Deep Learning. Cambridge, MA, USA: MIT Press, 2016.
- [15] J. Han, M. Kamber, and J. Pei, Data Mining: Concepts and Techniques, 3rd ed. Burlington, MA, USA: Morgan Kaufmann, 2011.
- [16] P. J. Rousseeuw and M. Hubert, "Anomaly detection by robust statistics," Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery, vol. 8, no. 2, pp. 1–20, 2018.
- [17] N. Japkowicz and S. Stephen, "The class imbalance problem: A systematic study," Intelligent Data Analysis, vol. 6, no. 5, pp. 429–449, 2002.
- [18] T. Fawcett, "An introduction to ROC analysis," Pattern Recognition Letters, vol. 27, no. 8, pp. 861–874, 2006.
- [19] J. Davis and M. Goadrich, "The relationship between Precision-Recall and ROC curves," in Proc. 23rd International Conference on Machine Learning (ICML), 2006, pp. 233–240. [20] G. E. A. P. A. Batista, R. C. Prati, and M. C. Monard, "A study of the behavior of several methods for balancing machine learning training data," SIGKDD Explorations, vol. 6, no. 1, pp. 20–29, 2004.

